Bittle X Pi Add-on v0.10 - Build Guide

Version 0.10, 2026-10-14. A sample add-on for the Bittle X platform v1.1 (../bittle_platform). It carries a Raspberry Pi Zero 2 W with a PiSugar 2 (1200 mAh), an NS4168 I2S amp with a 3 W speaker, and an SPH0645 I2S mic. It clicks onto the platform pegs. No tools, glue, tape or supports are necessary.



The add-on on the platform (lid in place).

1. Parts to print

File (stl/print/)	Material	Orientation	Supports	Cura profile	Mass
pod_tray.stl	PLA or PETG	floor on the bed	none	Bittle Pi Pod - PLA 0.25	12 g
pod_lid.stl	PLA	upside down: flat top on the bed	none	Bittle Pi Pod 0.25, or Bittle Fine 0.10 for the best surface	35 g
usb_plug.stl (optional)	TPU	flange on the bed	none	TPU, 20 mm/s	0.5 g
sw_slider.stl, sw_knob.stl	PLA	plate / knob face on the bed	none	Bittle Pi Pod 0.25	0.7 g
cover_pegs.stl (../bittle_platform)	PLA	stand on the rear end	tree, from the bed	Bittle Cover - PLA 0.25	11 g

Print quality. Every surface of the add-on parts is at 45 deg or less from vertical, or is a short bridge:

1. Tray: 6 mm² of overhang in total. These are the flat bottoms (1.0 wide) of the amp pins. The speaker pins have a 45 deg point at the bottom.
2. Lid: 75 mm² in total. These are the tops (in the print) of the charge window, the switch slot and the button slit: bridges of 4 to 7 mm. The profiles have the bridge settings on.
3. The mic holes in the left lid wall are diamonds, so they print clean without a bridge.
4. The slots for the mic board and the speaker ear plate are vertical in the print. They have no ceilings.
5. Keep the seam at the back (the profiles do this).

2. Electronics

Part	Notes
Raspberry Pi Zero 2 W	No header. Solder the wires to the pads, flat along the GPIO edge.
PiSugar 2, 1200 mAh	Under the Pi (pogo pins and its screws). The charge port, button and switch face the right side.
NULLLAB NS4168 I2S amp	38.4 x 22.4 mm board, on edge, crosswise behind the stack, on two pins. Components to the rear.
NULLLAB 3 W 4 ohm speaker	Ear plate 45 x 31 x 1.5, body 27 x 30 x 14.3 (ruler). On two pins at the front, sound to the front grille.

Part	Notes
HiLetgo SPH0645 I2S mic	16.7 x 12.7 x 1.8 board, on edge on the left side, beside the GPIO edge of the Pi, port to the left. Solder the wires at the top edge.
5 Dupont jumpers	Straight plugs on the BiBoard header (GND, +5V, TX2, RX2) in the cover slot.
Petoi AI Vision Module (optional)	Stays in the head. Its Grove cable goes along the left side of the robot and through the slot in the left lid wall to the Pi.

Balance. The add-on weighs about 119 g: the printed parts in PLA and typical masses of the electronics. Its center of mass is 9.4 mm behind the robot center, -0.0 mm right of the center line and 15 mm above the cover top. The speaker at the front balances the Pi stack. The masses are not weighed: put your own values in MASS_G in pod.py and run pod.py again.

3. Assembly

The 3D viewer shows each step with animation: https://beetlebugorg.github.io/bittlex/viewer/#step=1. Each step below has a link to its view.

1 Print the platform cover and fit it on the robot in place of the stock cover. (viewer step 2)

2 Solder the wires to the Pi, the amp and the mic (section 4).

3 Push the mic board down into the slot at the left side of the tray, port to the left, wire pads at the top. (viewer step 5)

4 Push the amp onto its two pins behind the stack, components to the rear. (viewer step 6)

5 Hang the speaker on the two pins of the front wall, ear plate to the rear. (viewer step 7)

6 Put the Pi + PiSugar 2 stack into the tray, battery down, ports to the right. (viewer step 8)

7 Put the tray on the cover 7 mm forward of its place, over the pegs. Push it 7 mm rearward until it clicks. (viewer step 3) (viewer step 4)

8 Push the Dupont plugs onto the BiBoard header. Route the wires over the Pi. (viewer step 9)

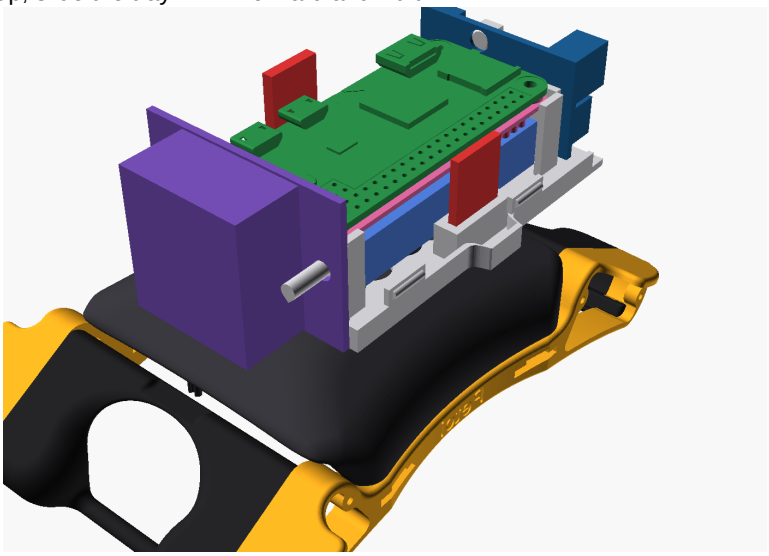
9 Optional camera: put the Grove cable through the slot in the left lid wall, behind the speaker. Solder its wires to the Pi (section 4). Leave slack for the head to turn.

10 Push the switch slider onto the stem of the knob through the switch slot of the lid: the knob outside, the fork down. It is a press fit.

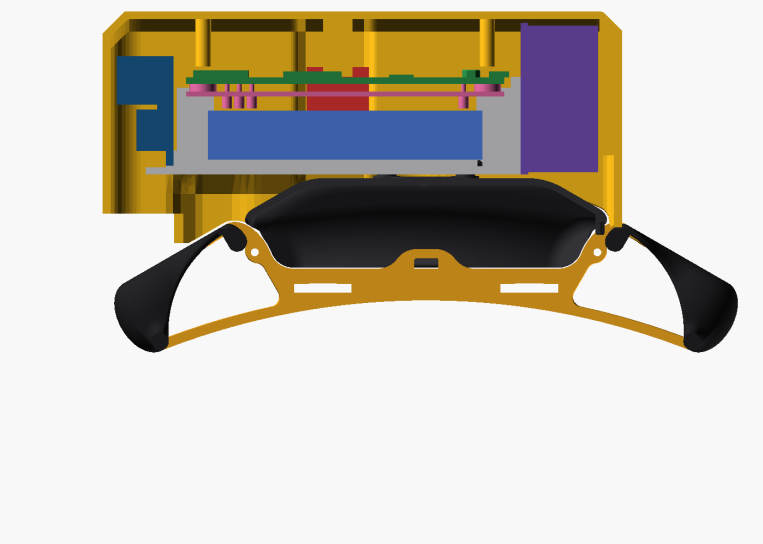
11 Set the knob to the position of the PiSugar switch. Push the lid down until the 4 snaps engage. The fork goes over the switch lever. The lid also stops the tray from sliding forward. (viewer step 10)

12 Optional: push the TPU plug into the charge window. (viewer step 11)

13 To remove: pull the lid up, slide the tray 7 mm forward and lift it.



The tray without the lid: the speaker (front), the stack on the center line, the mic (left) and the amp (rear).



Section on the center line (right: front). The speaker hangs on the front pins. The amp stands behind the stack.

4. Wiring

Signal	Pi Zero 2 W	To
UART TX / RX	GPIO14 pin 8 / GPIO15 pin 10	BiBoard RX2 / TX2
GND	pin 6	BiBoard GND
I2S BCLK	GPIO18 pin 12	amp BCLK and mic BCLK
I2S LRCLK	GPIO19 pin 35	amp LRCLK and mic LRCL
I2S out / in	GPIO21 pin 40 / GPIO20 pin 38	amp DIN / mic DOUT (mic SEL to GND)
Power	5V pin 2, 3V3 pin 1, GND pin 9	amp 5V, mic 3V. Do not connect BiBoard 5V to the Pi.
Speaker	-	amp + / - to the speaker
Camera I2C (Grove)	SDA GPIO2 pin 3, SCL GPIO3 pin 5, 3V3 pin 17, GND pin 14	AI Vision Module Grove: SDA, SCL, VCC, GND

5. Files

File	Content
stl/print/	the parts in print orientation
cad/pod_tray.step, cad/usb_plug.step, cad/sw_*.step	exact CAD (the lid is a mesh: its lower edge follows the robot)
cad/assembly_step.zip	one STEP file with the robot, the platform and the add-on in place, with named and colored parts. Unzip it and open it in FreeCAD or other CAD software
stl/installed/, stl/models/	the parts and the electronics in the robot frame
pod.py	all dimensions, the build and the checks (fit, slide path, loose pieces, overhang)
render.py, docs.py	the pictures and this guide

The PiSugar 2 ports are about 12 mm inside the right lid wall: the stack is on the center line for balance. Use a micro-USB cable with a plug body of 12 x 8 mm or less. The knob on the right side moves the switch: the fork on the slider holds the switch lever.

The lid closes the Pi ports. Remove the lid to use them. To remove the SD card, lift the stack out (the wires stay on): the amp is 0.2 mm behind the card.

Open items: the speaker sound face is assumed opposite the ears. The mic port is assumed at the board center; the 7 holes cover ±2.2 mm. The camera supply voltage is not checked: if the module needs 5 V, use pin 4. The PiSugar 2 LEDs are between the two boards and are not visible with the lid on. The BiBoard header height is an estimate.

6. License

The printed parts, the drawings and this guide are CC BY 4.0, and the code is MIT (Jeremy Collins). The Pi Zero and PiSugar 2 fit models are from github.com/PiSugar/PiSugar (GPL-3.0). The Petoi parts are from github.com/PetoiCamp/NonCodeFiles, without a stated license. LICENSE.md and ATTRIBUTION.md in the repository give the details.